PostgreSQL - Database & Table Operations

```
Server [localhost]:
Database [postgres]:
Port [5432]:
Username [postgres]:
Password for user postgres:

psql (18.1)
WARNING: Console code page (437) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \l
                                                      List of databases
   Name    |  Owner   | Encoding | Locale Provider |      Collate       |       Ctype        | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+--------------------+--------------------+--------+-----------+---------------------
--
 postgres  | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 student   | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 template0 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres
+
           |          |          |                 |                    |                    |        |           |
postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres
+
           |          |          |                 |                    |                    |        |           |
postgres=CTc/postgres
(4 rows)


postgres=# create database demodb;
CREATE DATABASE
postgres=# \l
                                                      List of databases
   Name    |  Owner   | Encoding | Locale Provider |      Collate       |       Ctype        | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+--------------------+--------------------+--------+-----------+---------------------
--
 demodb    | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 postgres  | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 student   | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 template0 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres
+
           |          |          |                 |                    |                    |        |           |
postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres
+
           |          |          |                 |                    |                    |        |           |
postgres=CTc/postgres
(5 rows)


postgres=# \c demodb
You are now connected to database "demodb" as user "postgres".
demodb=# create database test;
CREATE DATABASE
demodb=# \l
                                                      List of databases
   Name    |  Owner   | Encoding | Locale Provider |      Collate       |       Ctype        | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+--------------------+--------------------+--------+-----------+---------------------
--
 demodb    | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 postgres  | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 student   | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 template0 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres
+
           |          |          |                 |                    |                    |        |           |
postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres
+
           |          |          |                 |                    |                    |        |           |
postgres=CTc/postgres
 test      | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
(6 rows)


demodb=# drop database test;
DROP DATABASE
demodb=# \l
                                                      List of databases
   Name    |  Owner   | Encoding | Locale Provider |      Collate       |       Ctype        | Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+--------------------+--------------------+--------+-----------+---------------------
--
 demodb    | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 postgres  | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
 student   | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           |
```

```
 template0 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres
+
           |          |          |                 |                    |                    |        |           |
postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | English_India.1252 | English_India.1252 |        |           | =c/postgres
+
           |          |          |                 |                    |                    |        |           |
postgres=CTc/postgres
(5 rows)


demodb=# create database student;
ERROR:  database "student" already exists
demodb=# drop database student;
DROP DATABASE
demodb=# create database student;
CREATE DATABASE
demodb=# \c student
You are now connected to database "student" as user "postgres".
student=# CREATE TABLE students(name text,number int,age int);
CREATE TABLE
student=# \d
          List of relations
 Schema |   Name   | Type  |  Owner
--------+----------+-------+----------
 public | students | table | postgres
(1 row)


student=# INSERT INTO students(name,number,age) VALUES('Chidatma',27,18);
INSERT 0 1
student=# INSERT INTO students(name,number,age) VALUES('Rahul',38,20);
INSERT 0 1
student=# SELECT * FROM students;
   name   | number | age
----------+--------+-----
 Chidatma |     27 |  18
 Rahul    |     38 |  20
(2 rows)


student=# SELECT name FROM students;
   name
----------
 Chidatma
 Rahul
(2 rows)


student=# SELECT * FROM students WHERE number=27;
   name   | number | age
----------+--------+-----
 Chidatma |     27 |  18
(1 row)


student=# SELECT * FROM students WHERE age=20;
 name  | number | age
-------+--------+-----
 Rahul |     38 |  20
(1 row)


student=# SELECT number FROM students WHERE age=18;
 number
--------
     27
(1 row)


student=# SELECT number FROM students WHERE name='Chidatma';
 number
--------
     27
(1 row)


student=# SELECT name FROM students WHERE number=27;
   name
----------
 Chidatma
(1 row)


student=# TRUNCATE TABLE students;
TRUNCATE TABLE
student=# \d
          List of relations
 Schema |   Name   | Type  |  Owner
--------+----------+-------+----------
 public | students | table | postgres
(1 row)
```

```
student=# SELECT * FROM students;
 name | number | age
------+--------+-----
(0 rows)
```

CMD:

```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Chidatma>cd desktop

C:\Users\Chidatma\Desktop>cd Visual_studio

C:\Users\Chidatma\Desktop\Visual_studio>cd envcjp

C:\Users\Chidatma\Desktop\Visual_studio\envcjp>pip install psycopg2
Requirement already satisfied: psycopg2 in c:\users\chidatma\appdata\local\programs\python\python313\lib\site-packages (2.9.11)

[notice] A new release of pip is available: 25.2 -> 25.3
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\Chidatma\Desktop\Visual_studio\envcjp>cd scripts

C:\Users\Chidatma\Desktop\Visual_studio\envcjp\Scripts>activate

(envcjp) C:\Users\Chidatma\Desktop\Visual_studio\envcjp\Scripts>deactivate
C:\Users\Chidatma\Desktop\Visual_studio\envcjp\Scripts>cd..

C:\Users\Chidatma\Desktop\Visual_studio\envcjp>cd..

C:\Users\Chidatma\Desktop\Visual_studio>cd..
```

VS CODE:

```python
import psycopg2
conn = psycopg2.connect(dbname="postgres", user="postgres", password="Chidatma09", host="localhost", port=5432)
print("Connected Successfully")
```

```
PS C:\Users\Chidatma\Desktop\Visual_studio> python -u
"c:\Users\Chidatma\Desktop\Visual_studio\test.py"
Connected Successfully
```

```python
def table():
    conn = psycopg2.connect(dbname="postgres", user="postgres", password="Chidatma09", host="localhost", port=5432)
    cursor = conn.cursor()
    cursor.execute('create table employees (Name Text, ID int, Age int);''')
    print('Table created successfully')
    conn.commit()
    conn.close()
```

```python
def extract():
    conn = psycopg2.connect(dbname="postgres", user="postgres", password="Chidatma09", host="localhost", port=5432)
    cursor = conn.cursor()
    cursor.execute('''select * from employees;''')
    show = cursor.fetchone()
    print(show)
    print(show[1])
    print('Data added successfully  ')
    conn.commit()
    conn.close()
extract()
```

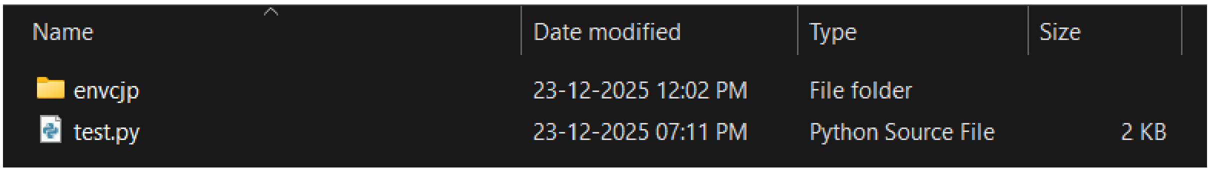
```
('Sam', 1, 30)
1
Data added successfully
```

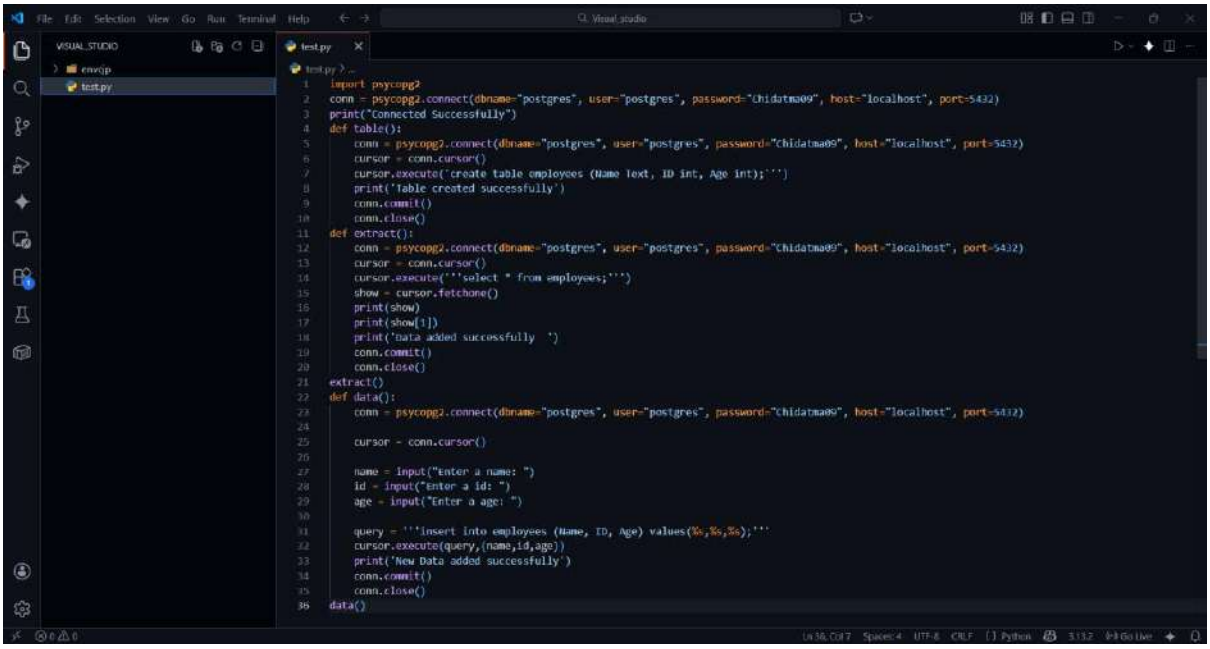
```python
def data():
    conn = psycopg2.connect(dbname="postgres", user="postgres", password="Chidatma09", host="localhost", port=5432)

    cursor = conn.cursor()

    name = input("Enter a name: ")
    id = input("Enter a id: ")
    age = input("Enter a age: ")

    query = '''insert into employees (Name, ID, Age) values(%s,%s,%s);'''
    cursor.execute(query,(name,id,age))
    print('New Data added successfully')
    conn.commit()
    conn.close()
data()
```

```
Enter a name: Chaitanya
Enter a id: 06
Enter a age: 30
New Data added successfully
```

PSQL:

```
Server [localhost]:
Database [postgres]:
Port [5432]:
Username [postgres]:
Password for user postgres:

psql (18.1)
WARNING: Console code page (437) differs from Windows code page (1252)
         8-bit characters might not work correctly. See psql reference
         page "Notes for Windows users" for details.
Type "help" for help.

postgres=# \d
           List of relations
 Schema |   Name    | Type  |  Owner
--------+-----------+-------+----------
 public | employees | table | postgres
 public | students  | table | postgres
(2 rows)


postgres=# select * from employees;
      name      |   id   | age
----------------+--------+-----
 Sam            |      1 |  30
 Chidatma Patel | 200709 |  18
 Para           | 200318 |  22
 Mitra          |   2015 |  11
 Chidatma       |     27 |  18
 Chidatma       |     27 |  18
(6 rows)


postgres=# \D
invalid command \D
Try \? for help.
postgres=# \d
           List of relations
 Schema |   Name    | Type  |  Owner
--------+-----------+-------+----------
 public | employees | table | postgres
 public | students  | table | postgres
(2 rows)


postgres=# select * from employees;
      name      |   id   | age
----------------+--------+-----
 Sam            |      1 |  30
 Chidatma Patel | 200709 |  18
 Para           | 200318 |  22
 Mitra          |   2015 |  11
 Chidatma       |     27 |  18
 Chidatma       |     27 |  18
 Chaitanya      |      6 |  30
(7 rows)
```


Name
Date modified
Type
Size
envcjp
23-12-2025 12:02 PM
File folder
test.py
23-12-2025 07:11 PM
Python Source File
2 KB


File Edit Selection View Go Run Terminal Help
Visual_studio
VISUAL_STUDIO
envcjp
test.py
import psycopg2
conn = psycopg2.connect(dbname="postgres", user="postgres", password="Chidatma09", host="localhost", port=5432)
print("Connected Successfully")
def table():
    conn = psycopg2.connect(dbname="postgres", user="postgres", password="Chidatma09", host="localhost", port=5432)
    cursor = conn.cursor()
    cursor.execute('''create table employees (Name Text, ID int, Age int);''')
    print('Table created successfully')
    conn.commit()
    conn.close()
def extract():
    conn = psycopg2.connect(dbname="postgres", user="postgres", password="Chidatma09", host="localhost", port=5432)
    cursor = conn.cursor()
    cursor.execute('''select * from employees;''')
    show = cursor.fetchone()
    print(show)
    print(show[1])
    print('Data added successfully ')
    conn.commit()
    conn.close()
extract()
def data():
    conn = psycopg2.connect(dbname="postgres", user="postgres", password="Chidatma09", host="localhost", port=5432)
    cursor = conn.cursor()
    name = input("Enter a name: ")
    id = input("Enter a id: ")
    age = input("Enter a age: ")
    query = '''insert into employees (Name, ID, Age) values(%s,%s,%s);'''
    cursor.execute(query,(name,id,age))
    print('New Data added successfully')
    conn.commit()
    conn.close()
data()
Ln 36, Col 7 Spaces: 4 UTF-8 CRLF {} Python